TrES-2b

R-1424

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_130619` · created 2026-08-01T06:37:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456462.83708 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.075 +/- 0.04
Transit depth (Rp/Rs)^2	0.56 +/- 0.6 [%]
Orbital Inclination (inc)	83.46 +/- 1.76
Airmass coefficient 1 (a1)	1.0123 +/- 0.0072
Airmass coefficient 2 (a2)	-0.011 +/- 0.026
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	#3 - [426, 152]
Optimal Method	PSF photometry
Transit Duration (day)	0.051 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.075 ± 0.04 vs 0.1254 (deviation 1.22σ)
Duration fit vs published	0.051 d vs 0.0746 d (deviation 0.67σ)
Mid-time O–C	-0.6 min
Depth SNR	1.8
Residual scatter	0.71 %

Light curve

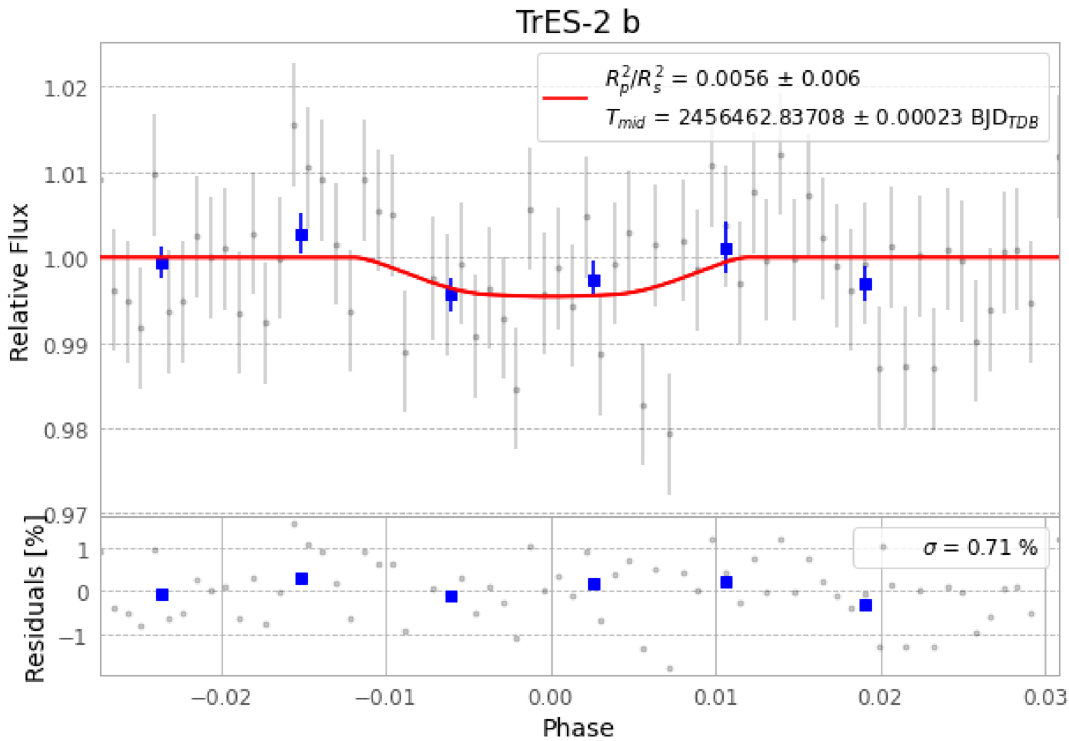

Figure 1 — FinalLightCurve_TrES-2 b_2013-06-18.png (SHA-256 951129d0f9a87c5c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 217], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a03d1101ca036c5a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

70 FITS frames (46 MB), `TRES-2130619062412.FITS` → `TRES-2130619095112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130619.log (c6d65c21ddd3…), FinalLightCurve_TrES-2 b_2013-06-18.png (951129d0f9a8…), FinalLightCurve_TrES-2 b_2013-06-18.csv (c90a8f18498c…)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)